

SİBER GÜVENLİK: KAPSAMLI REHBER VE UYGULAMA KILAVUZU

İÇİNDEKİLER

BÖLÜM 1: BİLGİ GÜVENLİĞİ TEMELLERİ VE VERİ KORUMA STRATEJİLERİ

Sayfa: 1-85

1.1 Bilgi Güvenliğinin Temel Kavramları ve İlkeleri

- CIA Triad: Confidentiality, Integrity, Availability
- Bilgi Güvenliği Risk Yönetimi ve Metodolojileri
- Güvenlik Görevleri Ayrılığı (Segregation of Duties)
- Savunma Derinliği (Defense-in-Depth) Stratejisi
- Need-to-Know ve Least Privilege Prensipleri

1.2 Bilgi Güvenliği Standartları ve Framework'ler

- ISO 27001/27002 Information Security Management
- NIST Cybersecurity Framework (CSF) Uygulamaları
- COBIT 5 IT Governance Framework
- COSO Internal Control Framework
- PCI DSS Payment Card Industry Standards

1.3 Veri Sınıflandırması ve Yaşam Döngüsü Yönetimi

- Veri Kategorileri: Public, Internal, Confidential, Restricted
- Data Ownership ve Data Stewardship Modelleri
- Veri Yaşam Döngüsü: Creation, Storage, Use, Share, Archive, Destroy
- Metadata Yönetimi ve Otomatik Sınıflandırma Araçları
- Veri Retention ve Disposal Politikaları

1.4 Şifreleme Teknolojileri ve Key Management

- Simetrik ve Asimetrik Şifreleme Algoritmaları (AES, RSA, ECC)
- Hash Fonksiyonları ve Digital Signatures
- Public Key Infrastructure (PKI) ve Certificate Management
- Hardware Security Modules (HSM) ve Key Management Systems (KMS)
- Quantum-safe Cryptography ve Post-Quantum Algorithms

1.5 Data Loss Prevention (DLP) ve Veri Koruma Teknolojileri

- Network-based, Endpoint-based ve Storage-based DLP
- Content Inspection ve Pattern Matching Teknikleri
- Data Classification Integration ve Policy Enforcement
- Cloud DLP ve Multi-cloud Veri Koruma Stratejileri
- Veri Maskeleyme (Data Masking) ve Anonimleştirme Teknikleri

1.6 Compliance ve Regulatory Frameworks

- GDPR (General Data Protection Regulation) ve Privacy by Design
 - KVKK (Kişisel Verilerin Korunması Kanunu) Uygulamaları
 - HIPAA Healthcare Information Privacy
 - SOX (Sarbanes-Oxley) Financial Data Protection
 - Industry-specific Compliance Requirements (FISMA, FERPA)
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BÖLÜM 2: AĞ GÜVENLİĞİ MİMARİSİ VE ALTYAPI KORUMA

Sayfa: 86-170

2.1 Ağ Güvenlik Mimarisi ve Tasarım Prensipleri

- Defense-in-Depth Network Architecture
- Network Segmentation ve Micro-segmentation Stratejileri
- Zero Trust Network Architecture (ZTNA) Modeli
- Software-Defined Perimeter (SDP) Yaklaşımları
- DMZ (Demilitarized Zone) Tasarımı ve Best Practices

2.2 Next-Generation Firewall (NGFW) ve Güvenlik Duvarları

- Stateful Packet Inspection vs Deep Packet Inspection
- Application-aware Firewalling ve Layer 7 Security
- Intrusion Prevention System (IPS) Integration
- SSL/TLS Decryption ve Content Filtering
- Firewall Rule Optimization ve Policy Management

2.3 Network Intrusion Detection ve Prevention Systems

- Signature-based vs Anomaly-based Detection Methods
- Network Behavior Analysis (NBA) Teknikleri
- Threat Intelligence Integration ve IOC Matching
- False Positive Reduction ve Tuning Strategies
- Distributed IDS/IPS Deployment Architectures

2.4 Secure Remote Access ve VPN Teknolojileri

- IPSec VPN: Site-to-Site ve Remote Access Configurations
- SSL/TLS VPN ve Web-based Remote Access
- Software-Defined WAN (SD-WAN) Security
- Zero Trust Network Access (ZTNA) Platforms
- Mobile VPN ve BYOD Security Considerations

2.5 Wireless Network Security ve 802.11 Protokolleri

- WPA3 ve Enterprise Wireless Security (802.1X/EAP)
- Wireless Intrusion Detection Systems (WIDS)
- Rogue Access Point Detection ve Mitigation
- Guest Network Isolation ve Captive Portal Security
- IoT Device Wireless Security Challenges

2.6 Network Monitoring, Analysis ve Forensics

- Network Traffic Analysis (NTA) ve Flow Monitoring
- SIEM Integration ve Log Correlation
- Packet Capture ve Deep Packet Analysis
- Network Forensics Methodologies
- Bandwidth Management ve QoS Security Implications

BÖLÜM 3: ENDPOINT SECURITY VE SİSTEM KORUMA STRATEJİLERİ

Sayfa: 171-255

3.1 Endpoint Protection Platform (EPP) Technologies

- Next-Generation Antivirus (NGAV) ve Machine Learning
- Behavioral Analysis ve Heuristic Detection
- Application Control ve Software Restriction Policies
- Device Control ve Removable Media Protection
- Cloud-based vs On-premises EPP Deployment Models

3.2 Endpoint Detection and Response (EDR) Solutions

- Advanced Threat Detection ve Hunting Capabilities
- Forensic Data Collection ve Analysis
- Automated Response ve Remediation Actions
- Threat Intelligence Integration ve IOC Matching
- EDR Data Retention ve Compliance Considerations

3.3 Extended Detection and Response (XDR) Platforms

- Multi-vector Threat Detection ve Correlation
- Cross-platform Visibility: Endpoint, Network, Cloud, Email
- AI/ML-powered Analytics ve Automated Response
- SOAR Integration ve Orchestrated Response
- XDR Vendor Ecosystem ve Integration Challenges

3.4 Operating System Security ve Hardening

- Windows Security Features ve Configuration Hardening
- Linux Security: SELinux, AppArmor, ve Container Security
- macOS Security Architecture ve Enterprise Management
- Mobile Operating Systems: iOS/Android Security Models
- Virtualization Security: Hypervisor ve VM Isolation

3.5 System Configuration Management ve Compliance

- Security Baseline Configuration Standards (CIS Benchmarks)
- Group Policy Management ve Security Templates
- Configuration Management Tools (Ansible, Puppet, Chef)
- Continuous Compliance Monitoring ve Drift Detection
- Patch Management Automation ve Testing Procedures

3.6 Mobile Device Management (MDM) ve BYOD Security

- Enterprise Mobility Management (EMM) Solutions
 - Mobile Application Management (MAM) Strategies
 - Containerization ve Work Profile Management
 - Mobile Threat Defense (MTD) Integration
 - Remote Wipe ve Data Leakage Prevention
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BÖLÜM 4: UYGULAMA GÜVENLİĞİ VE SECURE SOFTWARE DEVELOPMENT

Sayfa: 256-340

4.1 Secure Software Development Lifecycle (SSDLC)

- Security by Design ve Shift-Left Security Approaches
- Threat Modeling Methodologies (STRIDE, PASTA, DREAD)
- Secure Coding Standards ve Best Practices
- Security Code Review ve Static Analysis Integration
- Security Testing Integration in CI/CD Pipelines

4.2 Web Application Security ve OWASP Framework

- OWASP Top 10 Vulnerabilities ve Mitigation Strategies
- Input Validation, Output Encoding ve Parameterized Queries
- Authentication ve Session Management Security
- Cross-Site Scripting (XSS) ve Cross-Site Request Forgery (CSRF)
- Content Security Policy (CSP) ve Security Headers

4.3 API Security ve Microservices Architecture

- RESTful API Security Best Practices
- OAuth 2.0, OpenID Connect ve JWT Token Security
- API Gateway Security Features ve Rate Limiting
- GraphQL Security Considerations
- Microservices Communication Security (mTLS, Service Mesh)

4.4 Application Security Testing (AST) Methodologies

- Static Application Security Testing (SAST) Tools
- Dynamic Application Security Testing (DAST) Automation
- Interactive Application Security Testing (IAST) Benefits
- Runtime Application Self-Protection (RASP) Implementation
- Software Composition Analysis (SCA) ve Open Source Security

4.5 DevSecOps Integration ve CI/CD Pipeline Security

- Security Tool Chain Integration ve Automation
- Container Security Scanning ve Image Vulnerability Management
- Infrastructure as Code (IaC) Security Scanning
- Security Gates ve Quality Control in DevOps
- Shift-Left Security Culture ve Developer Training

4.6 Mobile Application Security

- iOS Application Security: Code Signing, App Transport Security
 - Android Application Security: ProGuard, Certificate Pinning
 - Mobile App Penetration Testing Methodologies
 - Binary Protection ve Anti-tampering Techniques
 - Mobile Backend API Security Considerations
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BÖLÜM 5: BULUT GÜVENLİĞİ VE MODERN İŞ YÜKLERİ KORUMA

Sayfa: 341-425

5.1 Cloud Security Architecture ve Shared Responsibility Model

- IaaS, PaaS, SaaS Security Responsibilities
- Multi-cloud ve Hybrid Cloud Security Strategies
- Cloud Security Reference Architectures
- Cloud Security Alliance (CSA) Framework
- NIST Cloud Security Guidelines Implementation

5.2 Cloud Infrastructure Security ve Governance

- Infrastructure as Code (IaC) Security Best Practices
- Cloud Security Posture Management (CSPM) Tools
- Cloud Workload Protection Platform (CWPP) Solutions
- Cloud Access Security Broker (CASB) Implementation
- Multi-cloud Security Management ve Governance

5.3 Container ve Kubernetes Security

- Container Image Security ve Vulnerability Scanning
- Container Runtime Security ve Behavioral Monitoring
- Kubernetes Security: RBAC, Network Policies, Pod Security
- Container Orchestration Security Best Practices
- Service Mesh Security (Istio, Linkerd) Implementation

5.4 Serverless Security ve Function-as-a-Service (FaaS)

- Serverless Architecture Security Challenges
- Function-level Security ve Code Injection Prevention
- Event-driven Security ve Trigger Validation
- Serverless Application Monitoring ve Logging
- Third-party Integration Security in Serverless

5.5 Cloud Identity ve Access Management (CIAM)

- Cloud-native Identity Providers (AWS IAM, Azure AD, GCP IAM)
- Federated Identity Management ve Single Sign-On (SSO)
- Privileged Access Management (PAM) in Cloud Environments
- Identity Governance ve Automated Provisioning
- Multi-cloud Identity Integration Challenges

5.6 Cloud Data Protection ve Encryption

- Cloud Data Classification ve Labeling Strategies
 - Cloud Encryption: At-rest, In-transit, In-use
 - Cloud Key Management Service (KMS) Implementation
 - Database Encryption ve Transparent Data Encryption (TDE)
 - Backup Encryption ve Disaster Recovery Security
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BÖLÜM 6: DONANIM GÜVENLİĞİ VE FİZİKSEL KORUMA SİSTEMLERİ

Sayfa: 426-510

6.1 Hardware Security Fundamentals ve Root of Trust

- Hardware Security Module (HSM) Architecture
- Trusted Platform Module (TPM) ve Measured Boot
- Hardware Root of Trust ve Secure Boot Process
- Chip-level Security Features ve Secure Enclaves
- Supply Chain Hardware Security ve Counterfeit Detection

6.2 Firmware Security ve Low-level System Protection

- UEFI/BIOS Security ve Secure Boot Implementation
- Firmware Attestation ve Integrity Verification
- Bootloader Security ve Chain of Trust
- Embedded System Firmware Security
- Firmware Update Security ve Over-the-Air (OTA) Updates

6.3 Physical Security Controls ve Access Management

- Biometric Access Control Systems
- Smart Card ve RFID Security Technologies
- Physical Intrusion Detection Systems
- Video Surveillance ve Analytics Integration
- Environmental Monitoring ve Tamper Detection

6.4 Data Center ve Critical Infrastructure Security

- Data Center Physical Security Architecture
- Power ve Cooling Infrastructure Security
- Fire Suppression ve Emergency Response Systems
- Visitor Management ve Escort Procedures
- Physical Asset Tracking ve Inventory Management

6.5 IoT ve Embedded Device Security

- IoT Device Authentication ve Identity Management
- Secure Communication Protocols for IoT (MQTT, CoAP)
- Edge Computing Security Architecture
- Industrial IoT (IIoT) Security Considerations
- IoT Device Lifecycle Management ve Security Updates

6.6 Mobile Device Hardware Security

- Secure Element (SE) Technology in Mobile Devices
- Hardware Security Keys ve FIDO2 Implementation
- Biometric Authentication Security (Fingerprint, Face Recognition)
- Mobile Device Tamper Resistance ve Anti-debugging
- Hardware-based Mobile Device Management (MDM) Features

BÖLÜM 7: KİMLİK VE ERİŞİM YÖNETİMİ (IAM) SİSTEMLERİ

Sayfa: 511-595

7.1 Identity Management Fundamentals ve Architecture

- Digital Identity Lifecycle Management
- Identity Governance ve Administration (IGA) Framework
- Identity as a Service (IDaaS) vs On-premises Solutions
- Directory Services: Active Directory, LDAP, Cloud Directories
- Identity Federation ve Trust Relationships

7.2 Authentication Technologies ve Multi-Factor Authentication

- Password Policies ve Passwordless Authentication
- Multi-Factor Authentication (MFA) Methods ve Technologies
- Risk-based Authentication ve Adaptive Authentication
- Biometric Authentication Systems ve Accuracy Considerations
- Certificate-based Authentication ve Smart Card Integration

7.3 Authorization Models ve Access Control Frameworks

- Role-based Access Control (RBAC) Design ve Implementation
- Attribute-based Access Control (ABAC) Advanced Scenarios
- Policy-based Access Control ve Dynamic Authorization
- Discretionary vs Mandatory Access Control Models
- Zero Trust Access ve Continuous Authorization

7.4 Privileged Access Management (PAM) Solutions

- Privileged Account Discovery ve Inventory Management
- Password Vaulting, Rotation ve Session Management
- Just-in-Time (JIT) Access ve Temporary Privilege Elevation
- Privileged Session Monitoring ve Recording
- PAM Integration with Cloud ve DevOps Environments

7.5 Single Sign-On (SSO) ve Identity Federation

- SAML 2.0 Federation Implementation
- OAuth 2.0 ve OpenID Connect Protocols
- Web Access Management (WAM) Solutions
- Cross-domain Identity Management (SCIM) Protocol
- Social Identity Integration ve External Identity Providers

7.6 Identity Analytics ve User Behavior Monitoring

- User and Entity Behavior Analytics (UEBA) Implementation
 - Identity Risk Scoring ve Anomaly Detection
 - Access Certification ve Recertification Processes
 - Segregation of Duties (SoD) Monitoring ve Violation Detection
 - Identity Governance Reporting ve Compliance Dashboards
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BÖLÜM 8: SİBER TEHDİT İSTİHBARATI VE THREAT HUNTING

Sayfa: 596-680

8.1 Cyber Threat Intelligence (CTI) Fundamentals

- Threat Intelligence Lifecycle ve Collection Methods
- Strategic, Tactical, Technical ve Operational Intelligence
- Threat Actor Profiling ve Attribution Challenges
- Diamond Model ve Kill Chain Analysis
- Intelligence Requirements ve Priority Intelligence Requirements (PIR)

8.2 Threat Intelligence Platforms ve Standards

- MITRE ATT&CK Framework Integration
- STIX/TAXII Standards ve Information Sharing
- Threat Intelligence Platform (TIP) Selection ve Implementation
- Indicators of Compromise (IOC) Management
- Tactics, Techniques, and Procedures (TTP) Analysis

8.3 Threat Hunting Methodologies ve Techniques

- Hypothesis-driven Threat Hunting Approaches
- Data-driven Hunting ve Statistical Analysis
- Behavioral Analytics ve Anomaly Detection
- Hunt Team Organization ve Skill Development
- Threat Hunting Metrics ve Success Measurement

8.4 Advanced Persistent Threat (APT) Detection

- APT Lifecycle ve Long-term Persistence Techniques
- Lateral Movement Detection ve Analysis
- Living-off-the-Land Techniques Identification
- Command and Control (C2) Communication Analysis
- APT Attribution ve Threat Group Tracking

8.5 Malware Analysis ve Reverse Engineering

- Static Malware Analysis Techniques ve Tools
- Dynamic Analysis ve Sandbox Environments
- Behavioral Analysis ve Family Classification
- Anti-analysis Evasion Techniques
- Automated Malware Analysis ve YARA Rule Development

8.6 Threat Intelligence Integration ve Operationalization

- SIEM ve SOAR Platform Integration
 - Automated Threat Response ve Playbook Development
 - Intelligence Sharing Communities ve Trust Groups
 - Custom Threat Intelligence Development
 - Threat Intelligence ROI Measurement ve Effectiveness
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BÖLÜM 9: OLAY MÜDAHALE VE ADLİ BİLİŞİM SÜREÇLERİ

Sayfa: 681-765

9.1 Incident Response Framework ve Organization

- NIST SP 800-61 Incident Response Process
- Computer Security Incident Response Team (CSIRT) Structure
- Incident Classification, Prioritization ve Severity Assessment
- Communication Plans ve Stakeholder Management
- Legal, Regulatory ve Law Enforcement Coordination

9.2 Incident Detection, Analysis ve Investigation

- Security Event Monitoring ve Alert Correlation
- Initial Response ve Triage Procedures
- Evidence Collection ve Chain of Custody Management
- Root Cause Analysis ve Attack Vector Identification
- Timeline Construction ve Attack Reconstruction

9.3 Containment, Eradication ve Recovery Procedures

- Short-term ve Long-term Containment Strategies
- Network Isolation ve System Quarantine Techniques
- Malware Removal ve System Remediation Procedures
- Business Continuity ve Service Restoration
- Recovery Validation ve Security Posture Assessment

9.4 Digital Forensics Methodology ve Techniques

- Forensic Readiness ve Preparation Planning
- Evidence Acquisition: Live System ve Post-mortem Analysis
- Computer Forensics: Windows, Linux, ve macOS Investigation
- Network Forensics ve Traffic Analysis
- Mobile Device Forensics ve Cloud Forensics Challenges

9.5 Specialized Forensics ve Advanced Analysis

- Memory Forensics ve Volatile Data Analysis
- Database Forensics ve Application Log Analysis
- Industrial Control System (ICS/SCADA) Forensics
- Virtual Machine ve Container Forensics
- Cryptocurrency ve Blockchain Forensics

9.6 Post-Incident Activities ve Lessons Learned

- Incident Documentation ve Reporting Requirements
 - Lessons Learned Analysis ve Process Improvement
 - Threat Intelligence Development from Incidents
 - Training ve Awareness Program Updates
 - Legal Proceedings Support ve Expert Witness Testimony
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BÖLÜM 10: YÖNETİŞİM, RİSK YÖNETİMİ VE UYUMLULUK (GRC)

Sayfa: 766-850

10.1 Information Security Governance Framework

- Board-level Security Governance ve Oversight
- Information Security Strategy ve Policy Development
- Security Organization Structure ve Roles/Responsibilities
- Security Culture Development ve Awareness Programs
- Third-party Risk Governance ve Vendor Management

10.2 Enterprise Risk Management ve Cyber Risk

- ISO 31000 Risk Management Framework
- Cyber Risk Quantification ve Business Impact Analysis
- FAIR (Factor Analysis of Information Risk) Methodology
- Risk Appetite ve Tolerance Definition
- Risk Treatment Strategies ve Control Selection

10.3 Compliance Management ve Regulatory Frameworks

- Regulatory Compliance Assessment ve Gap Analysis
- Internal Audit Programs ve Control Testing
- External Audit Coordination ve Remediation Management
- Compliance Automation Tools ve Continuous Monitoring
- Cross-border Compliance ve Data Sovereignty

10.4 Security Metrics, KPIs ve Performance Measurement

- Security Performance Indicator Development
- Risk Metrics ve Trend Analysis
- Executive Dashboard Design ve Reporting
- Security Investment ROI Calculation
- Benchmark Analysis ve Peer Comparison

10.5 Business Continuity ve Disaster Recovery Planning

- Business Impact Analysis (BIA) ve Criticality Assessment
- Recovery Time Objective (RTO) ve Recovery Point Objective (RPO)
- Disaster Recovery Planning ve Testing
- Crisis Management ve Emergency Response
- Supply Chain Continuity ve Vendor Dependency Management

10.6 Privacy Management ve Data Protection Governance

- Privacy by Design ve Privacy Impact Assessments
- Data Protection Officer (DPO) Role ve Responsibilities
- Data Subject Rights Management ve Breach Notification
- Cross-border Data Transfer ve Adequacy Decisions
- Privacy Engineering ve Technical Controls Implementation

BÖLÜM 11: SIZMA TESTİ VE ETİK HACKING METODOLOJİLERİ

Sayfa: 851-935

11.1 Penetration Testing Frameworks ve Methodologies

- OWASP Testing Guide Implementation
- PTES (Penetration Testing Execution Standard) Process
- OSSTMM (Open Source Security Testing Methodology Manual)
- NIST SP 800-115 Technical Guide to Information Security Testing
- Penetration Testing Scoping ve Rules of Engagement

11.2 Information Gathering ve Reconnaissance

- Passive Information Gathering ve OSINT Techniques
- Active Reconnaissance ve Network Enumeration
- Social Media Intelligence (SOCMINT) Collection
- DNS Enumeration ve Subdomain Discovery
- Search Engine ve Public Database Mining

11.3 Vulnerability Assessment ve Exploitation

- Automated Vulnerability Scanning ve Manual Validation
- Exploit Development ve Proof-of-Concept Creation
- Web Application Penetration Testing Techniques
- Network Service Exploitation ve Post-exploitation
- Wireless Network Penetration Testing Methods

11.4 Social Engineering Testing ve Physical Security

- Social Engineering Campaign Design ve Execution
- Phishing Simulation ve Awareness Testing
- Physical Penetration Testing ve Facility Assessment
- OSINT-based Social Engineering Attack Vectors
- Human Factor Security Assessment Methods

11.5 Red Team Operations ve Advanced Persistent Threat Simulation

- Red Team vs Penetration Testing Differences
- Advanced Persistent Threat (APT) Simulation Campaigns
- Command and Control (C2) Infrastructure Setup
- Living-off-the-Land Techniques ve Evasion Methods
- Red Team Exercise Planning ve Execution

11.6 Bug Bounty Programs ve Responsible Disclosure

- Bug Bounty Program Structure ve Management
 - Researcher Communication ve Relationship Management
 - Vulnerability Validation ve Severity Assessment
 - Remediation Coordination ve Timeline Management
 - Legal Framework ve Researcher Protection
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BÖLÜM 12: MALWARE ANALİZİ VE TERSİNE MÜHENDİSLİK

Sayfa: 936-1020

12.1 Malware Analysis Fundamentals ve Classification

- Malware Types: Virus, Worm, Trojan, Ransomware, APT Malware
- Malware Analysis Environment Setup ve Isolation
- Static Analysis vs Dynamic Analysis Approaches
- Malware Sample Acquisition ve Safe Handling
- Analysis Tools Selection ve Workflow Development

12.2 Static Malware Analysis Techniques

- File Format Analysis ve PE/ELF Structure Examination
- String Analysis ve Artifact Extraction
- Import/Export Table Analysis
- Cryptographic Algorithm Identification
- Packer/Obfuscator Detection ve Unpacking Techniques

12.3 Dynamic Malware Analysis ve Behavior Monitoring

- Sandbox Environment Configuration ve Monitoring
- System Call Tracing ve API Monitoring
- Network Traffic Analysis ve C2 Communication
- File System ve Registry Monitoring
- Process ve Thread Behavior Analysis

12.4 Advanced Reverse Engineering Techniques

- Disassembly ve Assembly Language Analysis
- Debugger Usage ve Code Flow Analysis
- Anti-analysis Technique Detection ve Bypass
- Cryptographic Implementation Analysis
- Firmware ve Embedded System Reverse Engineering

12.5 Automated Analysis Systems ve Machine Learning

- Cuckoo Sandbox Deployment ve Customization
- YARA Rule Development ve Signature Creation
- Machine Learning in Malware Detection
- Behavioral Pattern Recognition ve Classification
- Threat Intelligence Integration ve IOC Generation

12.6 Malware Family Analysis ve Attribution

- Malware Family Clustering ve Classification
 - Code Reuse ve Shared Infrastructure Analysis
 - Attribution Techniques ve Threat Actor Profiling
 - Campaign Tracking ve Timeline Analysis
 - Collaboration with Security Community ve Researchers
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BÖLÜM 13: SOSYAL MÜHENDİSLİK VE İNSAN FAKTÖRÜ GÜVENLİĞİ

Sayfa: 1021-1105

13.1 İnsan Psikolojisi ve Sosyal Mühendislik Temelleri

- Cognitive Biases ve Decision-Making Vulnerabilities
- Social Psychology Principles in Security Context
- Trust Building ve Persuasion Techniques
- Authority, Scarcity, Reciprocity Influence Tactics
- Psychological Profiling ve Target Selection Methods

13.2 Digital Sosyal Mühendislik Saldırı Teknikleri

- Phishing ve Spear-phishing Campaign Design
- Business Email Compromise (BEC) ve CEO Fraud
- Vishing (Voice Phishing) ve Smishing (SMS Phishing)
- Social Media Manipulation ve Fake Profiles
- Deepfakes ve AI-generated Content in Social Engineering

13.3 Fiziksel Sosyal Mühendislik ve OSINT

- Tailgating, Shoulder Surfing ve Physical Infiltration
- Dumpster Diving ve Physical Information Gathering
- Badge Cloning ve Access Card Duplication
- Pretexting ve Phone-based Social Engineering
- OSINT Collection ve Social Media Reconnaissance

13.4 Organizational Social Engineering Vulnerabilities

- Employee Security Awareness Gap Assessment
- Insider Threat Indicators ve Behavioral Analysis
- Executive Targeting (Whaling) ve High-value Target Attacks
- Third-party ve Supply Chain Social Engineering
- Remote Work Environment Social Engineering Risks

13.5 Sosyal Mühendislik Savunma Stratejileri

- Security Awareness Training Program Design
- Phishing Simulation ve Measurement Programs
- Security Culture Development ve Behavioral Change
- Technical Controls: Email Filtering, Web Protection
- Incident Reporting ve Response Mechanisms

13.6 Advanced Social Engineering ve Future Threats

- AI-powered Social Engineering Attacks
 - Synthetic Identity Creation ve Manipulation
 - Influence Operations ve Disinformation Campaigns
 - Hybrid Warfare ve Nation-state Social Engineering
 - Counter-intelligence ve Defensive Social Engineering
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BÖLÜM 14: GÜVENLİK OPERASYONLARI VE SOC/NOC YÖNETİMİ

Sayfa: 1106-1190

14.1 Security Operations Center (SOC) Architecture

- SOC Organizational Models: In-house, Outsourced, Hybrid
- SOC Technology Stack ve Tool Integration
- SOC Roles ve Responsibilities (L1, L2, L3 Analysts)
- 24/7 Operations Management ve Shift Planning
- SOC Maturity Models ve Capability Assessment

14.2 Network Operations Center (NOC) ve SOC Integration

- NOC Functionality ve Network Monitoring Capabilities
- SOC-NOC Collaboration ve Information Sharing
- Unified Operations Center (UOC) Models
- Service Level Agreements (SLA) ve Performance Metrics
- Escalation Procedures ve Incident Handoff Processes

14.3 SIEM Platform Management ve Log Analytics

- SIEM Architecture Design ve Scalability Considerations
- Log Collection, Normalization ve Enrichment
- Correlation Rule Development ve Tuning
- Use Case Development ve Detection Engineering
- SIEM Performance Optimization ve Storage Management

14.4 Security Orchestration, Automation and Response (SOAR)

- SOAR Platform Selection ve Implementation
- Playbook Development ve Automated Workflows
- Security Tool Integration ve API Management
- Case Management ve Ticketing System Integration
- Automation ROI Measurement ve Process Optimization

14.5 Threat Detection Engineering ve Hunt Operations

- Detection Use Case Development Lifecycle
- Behavioral Analytics ve Anomaly Detection Rules
- Threat Hunting Program Implementation
- Hunt Hypothesis Development ve Validation
- Detection Coverage Assessment ve Gap Analysis

14.6 SOC Performance Management ve Continuous Improvement

- SOC Metrics Development: Efficiency, Effectiveness, Quality
- Alert Fatigue Reduction ve False Positive Management
- Analyst Training ve Skill Development Programs
- SOC Tool Consolidation ve Technology Roadmap
- Managed Security Service Provider (MSSP) Evaluation

14.7 Emerging Technologies in Security Operations

- AI/ML Integration in SOC Operations
 - User and Entity Behavior Analytics (UEBA) Implementation
 - Cloud-native Security Operations
 - DevSecOps Integration with Security Operations
 - Zero Trust Architecture Impact on SOC Operations
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EKLER

- **Ek A:** Uluslararası Güvenlik Standartları ve Certification Frameworks
- **Ek B:** Siber Güvenlik Araçları ve Teknoloji Matrisi
- **Ek C:** Compliance ve Audit Checklist Templates
- **Ek D:** Incident Response Playbooks ve Templates
- **Ek E:** Risk Assessment Methodologies ve Calculation Tools
- **Ek F:** Security Awareness Training Materials ve Resources
- **Ek G:** Threat Intelligence Sources ve Sharing Communities
- **Ek H:** Security Architecture Design Patterns ve Reference Models

KAYNAKLAR VE REFERANSLAR

DİZİN VE TERİMLER SÖZLÜĞÜ

YAZARLAR HAKKINDA

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